BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 152 · Lesson 152 · JavaScript: DOM, Events and Interactive Pages

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What symbol selects CSS class warning?

Q2. What does a cache hit mean?

Q3. When can binary search be used?

Q4. What data structure follows LIFO?

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

DOM — Document Object Model.

event — Action or change announced by browser.

handler — Code running in response to event.

textContent — Reads/writes text without parsing it as HTML.

New learning

Select node — querySelector locates a document element.

Listen — addEventListener registers a handler for a click.

Update safely — textContent changes the displayed text without parsing HTML.

Worked example

A button with id go is clicked. Set the text of paragraph #status to 'Ready'.

1. Select #go — Identify the input or diagram and predict the first result.

2. Register click listener — Follow the second stage and explain its importance.

3. Set #status.textContent — document.querySelector('#go').addEventListener('click', () => { document.querySelector('#status').textContent = 'Ready'; }); This waits for the click and updates plain text.

We Do · Practise with guidance

Read #name input after a #greet click and display 'Hello ' plus the name in #result.

Read #name.value

Identify the input or diagram and predict the first result.

Build greeting string

Follow the second stage and explain its importance.

Write safe text

Inside a click handler: const name=document.querySelector('#name').value; document.querySelector('#result').textContent='Hello '+name; textContent avoids interpreting the name as HTML.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Define DOM, event and event handler in context. [6]

Question 2

Write JS that reacts to clicking #send and sets #message to 'Sent'. [6]

Question 3

Explain two benefits of client-side DOM updates and why a server must not trust a client-side form check. [8]

Exit Ticket · Answer before checking

addEventListener('click', handler) immediately executes handler every time it is registered.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A dot, .warning.

Q2. Required data already present in cache.

Q3. On sorted data.

Q4. Stack.

You Do pages · Independent questions

Q1. DOM document object tree [2], event signal [2], function executed in response [2].

Q2. Select button [2], addEventListener click [2], set paragraph textContent [2].

Q3. Responsive feedback [2], less round-trip traffic [2], bypassable browser code [2], repeat on server [2].

Exit Ticket · Model answer

False; it registers a callback for a later click.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: